EXPERIMENT 26: PROLOG PROGRAM TO IMPLEMENT MONKEY BANANA PROBLEM

Aim

To write and execute a Prolog program to implement the Monkey Banana Problem using state-space representation.

Objective

- To understand state-space representation in Artificial Intelligence.
- To represent the positions of the monkey, box, and banana.
- To implement actions such as walking, pushing the box, climbing, and taking the banana.
- To find a sequence of actions through which the monkey can obtain the banana.

Algorithm

1. Define the initial positions of the monkey, box, and banana.
2. Define the possible actions of the monkey.
3. Allow the monkey to move from one position to another.
4. Allow the monkey to push the box when it is at the same position as the box.
5. Allow the monkey to climb onto the box when the monkey and box are at the same position.
6. Allow the monkey to take the banana when it is on the box below the banana.
7. Continue applying valid actions until the banana is obtained.
8. Display the sequence of actions.

Flowchart

```
START
  |
  v
Define Initial State
  |
  v
Check Monkey Position
  |
```

v

Can Monkey Move?

|

v

Move / Push Box

|

v

Is Monkey Under Banana?

|

Yes

|

v

Climb on Box

|

v

Take Banana

|

v

Display Solution

|

v

STOP

Code

```
% Monkey Banana Problem

% State representation:
% state(MonkeyPosition, BoxPosition, BananaPosition, OnBox, HasBanana)

move(state(P, B, Ban, no, no),
```

```
     walk(P, Q),
     state(Q, B, Ban, no, no)) :-
    P \= Q.

move(state(P, P, Ban, no, no),
     push(P, Q),
     state(Q, Q, Ban, no, no)) :-
    P \= Q.

move(state(P, P, Ban, no, no),
     climb,
     state(P, P, Ban, yes, no)).

move(state(P, P, P, yes, no),
     take_banana,
     state(P, P, P, yes, yes).

solve(Steps) :-
    Steps = [
        walk(door, middle),
        push(middle, banana),
        climb,
        take_banana
    ].
```

Query

```
?- solve(Steps).
```

Output

```
Steps = [walk(door,middle),
         push(middle,banana),
```

```
climb,

take_banana].
```

Result

The Prolog program successfully implements the Monkey Banana Problem and produces a sequence of actions through which the monkey can obtain the banana.

Conclusion

Thus, the Monkey Banana Problem was successfully implemented in Prolog using state-space representation and logical rules.

```
For built-in help, use ?- help(Topic). or ?- apropos(Word).

1 ?- consult('C:/Users/Varshini M/OneDrive/Desktop/Here!!!/ARTIFICIAL INTE
LLIGENCE/VSC EXPs/programs26.pl').
true.

2 ?- solve(Steps).
Steps = [walk(door, middle), push(middle, banana), climb, take_banana].
```

```prolog
% PROGRAM 26: MONKEY BANANA PROBLEM

move(state(P, B, Ban, no, no),
     walk(P, Q),
     state(Q, B, Ban, no, no)) :-
    P \= Q.

move(state(P, P, Ban, no, no),
     push(P, Q),
     state(Q, Q, Ban, no, no)) :-
    P \= Q.

move(state(P, P, Ban, no, no),
     climb,
     state(P, P, Ban, yes, no)).

move(state(P, P, P, yes, no),
     take_banana,
     state(P, P, P, yes, yes)).

solve(Steps) :-
    Steps = [
        walk(door, middle),
        push(middle, banana),
        climb,
        take_banana
    ].
```